Digital Task Intensity of Indian Occupations — results and status

Nair, Anand & Shome (Nirma University) · SAU Digital Markets · status at 22 October 2026, 07:30 IST
For co-authors. All numbers come from the result files in the project repository (`dti-india`).

> **In one paragraph.** The LLM instrument has scored all 3,450 NCO-2015 occupations and is stable and grounded in the official text. Linked to 466,095 PLFS wage earners (2017-18 to 2023-24), a one-standard-deviation higher India-specific DTI goes with **4.1 log points (≈ 4.2%) higher hourly wages** within the same occupational division (SE 1.3), somewhat below the 5.7 obtained with the US O*NET measure. The premium is concentrated at the top of the wage distribution and is absent for workers with at most primary education, informal regular jobs and casual work — the pattern the abstract calls a *digital-returns gap*. It is a conditional association, not a causal return. A second LLM ranks occupation groups almost identically (Spearman 0.91), and the DML estimate is similar across four ML learners. RQ2: the same LLM reads O*NET descriptions as about 2.5 times as digital as the NCO texts for matching occupations. Still to come: valid human validation.

1. What is running and what is left

Step	Status	Detail
Qwen2.5-14B scores all 3,450 NCO occupations × 5 runs	Done (21 Oct, 05:50)	17,250 / 17,250, 0 failed
Wage analysis with the NCO-LLM DTI (RQ3–RQ4)	Done (21 Oct, 08:34)	10 tables
Llama-3.1-8B scores all occupations (second model)	Done (21 Oct, 15:40)	17,250 / 17,250, 0 failed
Cross-model agreement; wage robustness with the Llama DTI	Done (21 Oct, 15:55)	Sections 2 and 3
DoubleML with four ML learners, occupation-clustered SEs	Done (21 Oct)	Section 4, last table
Qwen2.5-14B scores 923 O*NET occupations; India vs O*NET gap (RQ2)	Done (22 Oct, 01:04)	4,615 / 4,615, 0 failed
Human validation (coders)	Not usable yet	round 2 forms rejected, see Section 2

- Jobs run on PARAM Shavak (Quadro GP100) inside `screen`; they do not depend on anyone's laptop.
- All scoring and analysis is finished. Next: the paper draft is updated with these results. The human–LLM comparison runs as soon as valid coder forms arrive.
- Decisions still open for the professors: wage-model choices marked OPEN in `configs/analysis.yaml`; confirmation of the strict rubric (Section 2).

2. RQ1 — Is the LLM a reliable measurement instrument?

Set-up. Main instrument: Qwen2.5-14B-Instruct, strict rubric (score only digital work the NCO-2015 text names; silent → 0), five independent runs per occupation at temperature 0.7, median across runs. Five dimensions scored 0–4; DTI = equal-weight sum of the z-scored dimensions. The strict rubric was chosen in the pilot (50 occupations, two models, two rubric readings): the broader "described-task" reading barely reduced the share of all-zero occupations (48% vs 50%) and lowered agreement between models (κ 0.69 vs 0.83).

Stability across repeated runs (all 3,450 occupations)

Dimension	ICC(2,1)	Krippendorff α	All 5 runs identical	Occupations scored > 0
Basic digital operation	0.73	0.72	62%	33%
Digital information processing	0.82	0.81	72%	28%
Digital communication	0.70	0.65	90%	5%
Specialised digital tools	0.81	0.79	60%	36%
Digital problem solving / content	0.84	0.79	84%	13%

Meaning. ICC/α of 0.70–0.84 is "good" agreement by the usual benchmarks (0.75+ good, 0.6–0.75 moderate): asked five times, the model gives largely the same scores. Communication (D3) is the least stable because it is rarely non-zero, so a single deviating run weighs heavily. 1,190 occupations (34%) vary by more than 0.5 points (SD) on at least one dimension; they are listed for review, and the median across five runs is used precisely to absorb this noise.

Grounding and consistency

- **87%** of the 20,355 non-zero dimension scores come with a quote that is found in the official description (pilot: 89%). The rest are mostly scores without a quote, where the model drew on general knowledge despite the rule.
- Equal-weight dimension mean vs the model's own holistic 0–10 judgement (Q9): Spearman **0.84** — supports equal weighting.
- Correlation with a simple keyword count on the same text: Spearman **0.30**. Occupations whose description names a digital term average 1.56 (0–4) vs 0.39 for the rest; only 7% of the former are all-zero. The LLM therefore picks up much more than keywords, but stays tied to the text.
- **51.6%** of occupations score 0 on every dimension. This is a finding about the NCO-2015 text, not an error: only about 5% of descriptions use any digital term. It is the main reason to compare with O*NET (RQ2).

Mean score (0–4) by division

Division	Occ.	Mean	All-zero
1. Managers	180	0.30	68%
2. Professionals	647	1.05	31%
3. Technicians & assoc. professionals	546	0.65	37%
4. Clerical support	118	0.59	35%
5. Service & sales	154	0.28	66%
6. Skilled agriculture	119	0.05	89%
7. Craft & related trades	859	0.20	69%
8. Plant & machine operators	715	0.26	44%
9. Elementary occupations	112	0.06	88%

Most digital occupations

Occupation	Mean
Programmer Analyst	3.2
Software Engineer	3.2
Programmer, Engineering and Scientific/System Programmer	3.0
Astronomer	2.8
Caption Writer	2.8
Radio Engineer, Telecommunication	2.8
Web Designer	2.8
Associate-Analytics	2.6
Computer Programmer/Software Engineer	2.6
Computer Programmers, Other	2.6

Meaning. The ranking is economically sensible (professionals > technicians > clerks > … > agriculture and elementary work) and the most digital occupations are the expected ones — face validity for the instrument. PLFS match rate: 98.7–99.8% of employed workers in each round receive a DTI.

Agreement between two LLM families (Qwen2.5-14B vs Llama-3.1-8B, all 3,450 occupations)

Dimension	Exact	Within 1	Weighted κ	Spearman
Basic digital operation	69%	91%	0.51	0.55
Digital information processing	78%	92%	0.77	0.73
Digital communication	95%	98%	0.47	0.47
Specialised digital tools	72%	91%	0.62	0.63
Digital problem solving / content	87%	96%	0.78	0.72
Holistic score (Q9, 0–10)	36%	73%	0.73	0.70

Level	Units	Spearman, DTI
8-digit occupation	3450	0.73
4-digit family (NCO-2015)	373	0.89
3-digit group (NCO-2015, PLFS 2021-24)	122	0.91
3-digit group (NCO-2004, PLFS 2017-21)	111	0.86

Meaning. Two independently trained models, from different companies and of different size, rank occupation *groups* (the level at which DTI enters the wage regressions) almost identically (0.91 for the NCO-2015 groups). Agreement on single occupations is moderate, highest for information processing and content/problem solving and lowest for communication, which is rarely non-zero. Llama is the noisier instrument: run stability ICC 0.57–0.75 (Qwen 0.70–0.84) and 62% all-zero occupations (Qwen 52%). This supports Qwen as the main instrument.

> **Human validation not yet usable.** 200 occupations were sent to three coders (stratified by division and by whether the text names digital work; pilot occupations excluded). The two forms returned on 21 Oct failed the quality checks and are not used: one is a near-copy of the other (96% identical rows, identical evidence, same coder ID), no occupation was scored 0 although the rule requires 0 when the text is silent, agricultural and elementary jobs score like professionals, only 23% of evidence words appear in the description, and scores repeat in a fixed 10-row pattern. (The first round, 19 Oct, gave identical scores on all five dimensions.) A smaller sample coded independently, e.g. 100 occupations by two coders, would suffice.

3. RQ3 — Conditional digital occupational wage premium

ln(real hourly wage) of regular and casual wage earners aged 15–64, seven PLFS rounds, PLFS weights, standard errors clustered by occupation (233 occupation groups). Coefficients in **log points per one standard deviation of DTI** (≈ % difference in wages). SE in brackets; *** p<0.01, ** p<0.05, * p<0.1.

Specification	NCO-LLM DTI (India)	R^2	O*NET-transferred DTI
M1 No controls	27.5*** (3.2)	0.24	30.4*** (2.5)
M2 + individual characteristics	8.5*** (2.6)	0.46	11.2*** (1.9)
M3 + state, industry, year-quarter FE	6.3*** (1.3)	0.56	10.7*** (1.6)
M4 + occupational division FE (preferred)	4.1*** (1.3)	0.57	5.7*** (1.5)

Meaning. Without controls, more digital occupations pay 27.5 log points more per SD. Most of that reflects who works in them (education, age, location, industry). Comparing occupations within the same broad division, the premium is **4.1 log points (≈ 4.2%)**: modest but clearly different from zero (p = 0.002). With the US O*NET measure it is 5.7: transferring US task content gives a somewhat larger premium.

How robust is it?

Check (all M4)	NCO-LLM DTI	O*NET DTI
M4 unweighted	3.5*** (1.3)	6.4*** (1.6)
M4 unweighted, wild cluster bootstrap	3.5** (1.3)	6.4*** (1.6)
M4 monthly earnings	5.2*** (1.5)	8.4*** (1.4)
M4 regular wage workers only	3.9*** (1.3)	5.0*** (1.5)
M4 casual workers only	-1.6 (1.8)	0.9 (1.8)
M4 NCO-2015 rounds (Jul 2021+)	4.6** (2.1)	4.4*** (1.3)
M4 NCO-2004 rounds (pre Jul 2021)	3.1* (1.6)	6.4*** (2.0)
M4 NCO-2004 groups with ≥75% direct links	5.4*** (1.8)	5.5*** (1.5)
M4 self-employed (earnings incl. capital returns)	0.5 (1.1)	-3.9* (2.2)
Heckman selection correction (all)	4.2*** (1.3)	5.7*** (1.4)
DTI from the second LLM (Llama-3.1-8B)	2.2* (1.1)	—
DTI quintile 2 vs lowest fifth	3.9 (2.5)	0.3 (2.0)
DTI quintile 3 vs lowest fifth	1.5 (2.5)	2.5 (2.1)
DTI quintile 4 vs lowest fifth	8.3*** (2.8)	3.6 (2.5)
DTI quintile 5 vs lowest fifth	9.3*** (2.9)	13.5*** (4.1)

Meaning. The premium survives unweighted estimation, the wild cluster bootstrap, monthly earnings, direct links only and correction for selection into wage work; it is weaker (significant only at 10%) in the pre-2021 NCO-2004 rounds. With the DTI from the second LLM it is 2.2 ($p = 0.052$): same sign and pattern, smaller, as expected when the noisier Llama scores attenuate the estimate. It is **not linear**: only the two most digital fifths of occupations earn significantly more. It is absent for casual workers and the self-employed.

> **Not causal.** Oster (2019) $\delta = 0.34$ (O*NET: 0.40): selection on unobservables only about a third as strong as selection on observables would explain the premium away. Workers sort into digital occupations by ability and background, so the paper reports β as a **conditional occupational association**, as stated in the abstract.

4. RQ4 — Who gains? Heterogeneity

Across the wage distribution (unconditional quantile / RIF regressions)

Wage percentile	NCO-LLM DTI	O*NET DTI
10th percentile	-1.7 (1.3)	1.6 (1.5)
25th percentile	-0.2 (1.3)	1.8 (1.5)
50th percentile	2.1 (1.4)	3.6** (1.8)
75th percentile	6.2*** (1.7)	7.8*** (2.3)
90th percentile	21.3*** (6.7)	19.8*** (6.1)

Meaning. No premium in the lower half of the distribution; it rises to about 21.3 log points at the 90th percentile. Occupational digital intensity pays mainly at the top.

By group (M4 estimated separately for each group)

Group	NCO-LLM DTI	O*NET DTI
Education: up to primary	-2.1 (1.5)	-0.4 (2.3)
Education: middle / secondary	3.5*** (1.2)	4.8*** (1.5)
Education: tertiary	4.7*** (1.5)	7.3*** (1.8)
Regular, no social security (informal)	1.0 (0.8)	-0.3 (1.5)
Regular, with social security (formal)	3.8*** (1.2)	5.6*** (1.6)
Regular wage workers	3.9*** (1.3)	5.0*** (1.5)
Casual workers	-1.6 (1.8)	0.9 (1.8)
Men	3.6*** (1.2)	3.5** (1.4)
Women	5.6** (2.2)	8.6*** (2.0)
Urban	3.1** (1.3)	4.0*** (1.4)
Rural	4.6*** (1.7)	7.8*** (1.9)
Age 15–24	2.5** (1.1)	2.6** (1.2)
Age 45–54	4.3** (1.9)	8.3*** (2.2)

Within the same 3-digit occupation (interactions with full occupation fixed effects)

Difference in the premium	NCO-LLM DTI	O*NET DTI
Women vs men	3.7*** (0.7)	3.2*** (1.0)
Rural vs urban	-3.0*** (0.6)	-3.4*** (0.7)
Middle/secondary vs up to primary	8.6*** (2.4)	8.9*** (1.7)
Tertiary vs up to primary	11.5*** (3.3)	18.0*** (2.1)
Casual vs regular	-5.9** (2.7)	-0.2 (2.2)
Formal vs informal (regular workers)	4.0*** (1.4)	11.0*** (2.0)

Meaning.

- **Education is the clearest divide.** No premium for workers with at most primary schooling (-2.1, n.s.); 4.7 for graduates. Within the same occupation, graduates gain about 12 log points more per SD than the least educated.
- **Formality.** Among regular workers, the premium exists only with social-security coverage (3.8 vs 1.0, n.s.). Casual workers get none.
- **Gender.** Women earn a larger premium than men (5.6 vs 3.6; machine-learning group effects 6.3 vs 2.9), but occupational DTI explains almost none of the 31.7-point gender wage gap (Oaxaca–Blinder: 0.5 points; O*NET: -0.1). Most of the gap (28.0 points) is unexplained.
- **Rural–urban is mixed.** Estimated separately, the premium is higher for rural workers; within the same occupation it is lower. This needs care in the paper and is reported as inconclusive.
- **Machine-learning group effects** (double/debiased ML with gradient boosting) confirm the education and gender gradients.

Machine-learning check: does the result depend on the ML model?

Double/debiased ML (partially linear model, 200,000-worker random subsample, 5-fold cross-fitting by occupation, standard errors clustered by occupation). The same estimator is run with four learners for the two nuisance predictions (wage from controls, DTI from controls). R^2 = how well each learner predicts on held-out data.

Learner	Average effect (cluster SE)	95% CI	R^2, wage	R^2, DTI	O*NET DTI
Linear regression	4.0** (1.7)	[0.7, 7.4]	0.543	0.650	6.6*** (2.0)
Lasso (cross-validated)	2.6 (1.7)	[-0.7, 6.0]	0.545	0.651	6.9*** (1.8)
Random forest	4.4* (2.6)	[-0.8, 9.6]	0.504	0.703	3.9 (2.8)
Gradient boosting	3.3** (1.4)	[0.5, 6.1]	0.594	0.694	3.4 (2.1)

Meaning. The effect is 2.6–4.4 log points across learners, close to the regression estimate (4.1): the result does not hinge on one ML model. The best-fitting learner is gradient boosting (highest out-of-sample R^2 for wages), with 3.3 log points (95% CI 0.5 to 6.1). With occupation-clustered standard errors the intervals are much wider than in a DML run that treats workers as independent; with lasso, random forest the effect is not significant at 5%. With the O*NET-transferred DTI the spread is wider: 3.4–6.9 log points, and the best-fitting learner (gradient boosting) gives 3.4 ($p = 0.10$). The larger O*NET premium in the linear models therefore shrinks once controls enter flexibly, while the India-specific estimate is more stable across learners.

> **Policy reading (abstract, Section 7).** Digital intensity of an occupation translates into higher pay mainly for educated, formally employed, higher-paid workers. For the less educated, informal and casual workforce the premium is zero. This is a **digital-returns gap**, not only a digital-skills gap: basic digital training alone is unlikely to close it without formalisation, advanced skills and access to digitally enabled jobs.

5. RQ2 — India-specific vs O*NET-transferred intensity

The same LLM and prompt scored 923 O*NET occupations from their O*NET description and core tasks; both are compared at the NCO-2015 family level (361 families), mapped through SOC → ISCO-08 with US employment weights.

Do they rank occupations alike?

O*NET measure	Spearman
Digital-keyword share of core tasks	0.49
Working with Computers	0.61
Computers & Electronics knowledge	0.57
E-mail frequency	0.53
Muro et al. computer score	0.61
Same LLM prompt on O*NET text	0.68

Gap by division (same LLM, 0–4 scale)

Division	Fam.	O*NET – NCO
1. Managers	22	+1.40
2. Professionals	85	+1.03
3. Technicians & assoc. professionals	64	+0.90
4. Clerical support	25	+0.88
5. Service & sales	31	+0.27
6. Skilled agriculture	15	+0.90
7. Craft & related trades	61	+0.35
8. Plant & machine operators	37	+0.34
9. Elementary occupations	21	+0.02

Meaning. The two sources agree on *which* occupations are digital (Spearman 0.68 with the same instrument), so the India-specific DTI is externally anchored. But the *level* differs sharply: read by the same model with the same rubric, O*NET descriptions score 1.18 on average against 0.47 for the NCO-2015 texts (about 2.5 times as high; higher for 80% of families). The gap is largest for managers, professionals, technicians, clerks and skilled agriculture and close to zero for elementary work. **Answer to RQ2:** transferring US task content overstates the digital intensity of Indian occupations, especially white-collar and managerial work, and it gives a larger wage premium (5.7 vs 4.1) that is less robust to flexible ML adjustment. The comparison cannot by itself separate "Indian jobs are less digital" from "the 2015 NCO texts describe work in less digital terms"; the paper should say so.

6. Data and reproducibility

- NCO-2015 Vol II descriptions (3,450 occupations); PLFS 2017-18 to 2023-24 (seven annual rounds, CWS wages, MoSPI weights), pre-2021 rounds linked through an NCO-2015 → NCO-2004 concordance (2,657 direct links, others via siblings); O*NET 31.0 with BLS SOC–ISCO crosswalk and OEWS weights.
- LLMs run locally (llama.cpp, GGUF): no occupation text leaves the university server. Prompt and rubric are versioned (hash 46a323537a).
- PLFS microdata are not redistributed; only estimates appear in the repository. Code: private GitHub repository `dti-india` (guide for writing: `docs/PAPER_GUIDE.md`).

Preliminary results for internal discussion; numbers may change slightly once the human validation is added.